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RESEARCH ARTICLE

Enhancing Emoji-Based Sentiment Classification in Urdu Tweets: Fusion Strategies With Multilingual BERT and Emoji Embeddings

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ABSTRACT X (formerly known as Twitter) is a popular social network with hundreds of millions of users. We emphasize the benefits of using emojis to enhance the comprehension of user sentiment. Our objective was to analyze the sentiments expressed in Urdu language tweets, a task that can be demanding due to the language's intricate structure and diverse dialects. Our research revolves around combining emoji embeddings with the SentiUrdu-1M dataset, consisting of 1.14 million Urdu tweets and 1,194 emojis, using multilingual BERT (mBERT). The major motive of our study is twofold: 1) to evaluate the performance of pre-trained emoji2vec and our proposed method of Urdu-Specific FastText emoji embeddings in terms of their ability to distinguish emojis based on their expressions; and 2) to explore techniques for integrating Urdu tweets and emoji embeddings, including concatenation, neural network fusion, and attention mechanism fusion. Moreover, we fine-tuned the baseline models on only-text Urdu tweets using multilingual BERT and XLM-RoBERTa, achieving accuracies of 64% and 65%, respectively. Therefore, our study fills a gap in the literature by investigating the possibility of enhancing sentiment analysis in Urdu language tweets through emojis, a field that has received limited attention. The Urdu-Specific FastText emoji embeddings proposed in this paper yield better results than the pre-trained emojis from emoji2vec and improve sentiment analysis accuracy up to 95% for the neural network fusion approach.

INDEX TERMS Urdu tweets, sentiment analysis, fine-tuning, emojis, multilingual BERT, XLM-RoBERTa, emoji embeddings.

I. INTRODUCTION

In this digital era, social networks and microblogging are highly used form of sharing ideas, opinions, daily communication, reviews, or any viral news, as a results, organizations and institutes are searching for useful information from social media [1]. Twitter, which rebranded as X since July 31, 2023, has seen increased interest among researchers in recent years for developing techniques to detect sentiments in tweets posted on X (Twitter) [2]. These tweets can provide insight into people's emotions which are an essential part of the human life experience and are influential in individuals decision-making processes [3]. Urdu is the national language of Pakistan, which is also widely spoken in numerous areas of South Asia [4], [5] The value of the Urdu language lies in its script, which is the only exception in the contemporary world to employ a right-to-left orientation [6] and [7]. Analyzing sentiments in Urdu is just as significant as in any other language [8]. Many individuals who are proficient in Urdu utilize the language for expressing their thoughts and emotions on social media platforms such as X (Twitter), Facebook, and YouTube [9], [10]. It is critical to analyze Urdu

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text to understand the perceptions and attitudes of individuals who speak the Urdu language [11].

Emojis offer significant indications of a user's sentiments when analyzing short, casual comments, such as tweets [12]. The use of emojis enables us to analyze how individuals articulate their emotions in written comments [13]. The limited availability of data and resources for Urdu has resulted in less research on sentiment analysis for the language, specifically considering emojis in Urdu tweets is first work to our best knowledge. The presence of emojis intensifies this disparity, as they are widely used but have not been thoroughly examined in the context of understanding emotions in Urdu [14].

TABLE 1. Examples of dataset for urdu tweets with Emojis.

Urdu Tweets with Emojis	Label	English Translation
لگتا ہے اس فرضی ڈاکٹر کا ذہنی توازن کا دیوالیہ نکل گیا ہے😂	Neutral	It seems this fake doctor has gone bankrupt mentally 😂
شکریہ ❤️❤️	Love	Thank you ❤️❤️
کامران صاحب جو آپ کہہ رہے ہیں وہ اور بات ہے اور میری توقعات😅	Excitement	What you're saying, Mr. Kamran, is a different matter, and my expectations 😅
چالیس لوگوں کو جمع کر کیا میسج دینا چاہ رہے ہو؟🤔	Neutral	What message are you trying to convey by gathering forty people? 🤔
یقین کرو کوئی کسی کا نہیں ہوتا، دل کے دو حرف😔	Anger	Believe it, no one belongs to anyone, just the two letters of the heart 😔

Table 1 presents selected examples from our dataset, showcasing the integration of Urdu tweet with emojis and their corresponding sentiments, alongside English translations. These examples illustrate the nuanced way in which emojis can clarify the emotional tone of messages in digital Urdu discourse, highlighting the importance of considering both textual and emotive elements in sentiment analysis. Our research embarked upon here is poised to make several significant contributions to the field of sentiment analysis, particularly within the context of the Urdu language. The main contributions of this paper are as follows:

1. At the core of this research is the development of our unique emoji embedding approach named Urdu-Specific FastText emoji embeddings. Furthermore, we compared our proposed embedding technique with existing pre-trained emoji2vec embeddings. The Urdu-Specific FastText emoji embeddings designed specifically for the Urdu language, handle emojis in their alignment with Urdu description. This method is developed to enhance the sentiment and cultural dimensions of Urdu tweets, consequently, emojis are used tangibly to increase sentiment analysis richness.

2. Investigating the fusion strategies of Urdu tweets and emoji embedded representations, which establish a core part of our proposed research. We used three different fusion strategies: concatenation, fusion of neural networks and attention mechanism. The aim of our research is consequently finding the combination of textual and emoji-related sentiment elements. Our findings are anticipated to provide the community with new insights into sentiment analysis methods, which can be applied not only to Urdu language but also any other languages that may need an integration strategy for different input modalities.

3. Based on the carefully conducted experiments using the massive SentiUrdu-1M dataset, this research work delivers vital insights into the operational pattern of proposed embedding models and their fusion strategies. These insights contributes to the understanding of the role of emojis in the sentiment analysis of Urdu tweets, thereby laying the foundation for the development of more refined, context-sensitive sentiment analysis models [15].

4. Baseline results were obtained from only-text Urdu tweets using multilingual BERT (mBERT) and XLM-RoBERTa (XLM-R), as well as proposed emoji embedding and fusion techniques.

Foundational works, which present the contextual Urdu text emotion detection corpus [16], have cleared the ground for using advanced deep learning architectures for the sentiment analysis of Urdu text. In parallel, research on the role of emojis in the digital communication, such as studies on the classification of emojis using Siamese network architecture [17], has revealed the possibility that these symbols help to provide more emotional context of the textual data.

The structure of this research paper is organized as follows: Section II contains a literature review covering all research work related to the current study. Section III describes the dataset preparation step by step. Section IV discusses methods of handling emojis, highlighting the practical use of these techniques. Section V provides a detailed result of the experiments that were carried out, and Section VI concludes the paper with the key findings and outlines avenues for future investigation in sentiment analysis.

II. RELATED WORK

The domain of sentiment analysis in Urdu, a language with rich contextual and cultural complexities, has been significantly enhanced by the advent of smart embedding techniques and fusion strategies. This in-depth literature review attempts to reconcile the findings of various pivotal studies, specifically by examining the intricate relationship between different embedding strategies, particularly emoji2vec and Urdu-Specific FastText emoji embeddings, and fusion methods, such as concatenation, neural network fusion, and attention mechanism fusion, in effectiveness of sentiment analysis for Urdu text.

A. URDU SENTIMENT ANALYSIS USING TWEETS DATASET

The research by [18] identified sentiment in form of binary, positive and negative and classified emotions in positive, negative, and neutral forms using Markov chains. The research utilized 3,103 tweets as its dataset. The article compared lexical and machine learning based algorithms, revealing the suggested method had the greatest accuracy, 69% and 86.5% for ternary and binary sentiment classification, respectively. Brief research was conducted by [19] on Urdu tweets related to news articles for sentiment analysis. The study describes the data collection approach for 600 tweets, with 500 used to train the prediction algorithm. The data classification was done using only the decision tree classifier. A small dataset and a lack of comparison with other methods can cause the researchers to argue the recorded 90% accuracy of the experiment. The research by [20] on deep neural network-based sarcasm detection tweets used the commenter's past posts to determine context. The proposed convolutional neural network-based architecture achieved an accuracy of 63% using 1,500 datasets and 6,774 historical tracks. Another research was conducted on Urdu tweets dataset by [21] focused on sentiment analysis, focusing on political tweets from top Pakistani politicians in recent years. Machine learning classifiers, including Decision Trees, SVM, and XGBoost, were used in the experiment to predict the sentiment polarity. The study also revealed several challenges, such as issues with imbalanced datasets and the limitations of models like Word2Vec in capturing the nuances of the Urdu language. The results emphasize the importance of custom deep learning models for sentiment analysis from Urdu textual data, particularly in a political context. Reference [22] extended this work by analyzing sentiment in Urdu using data from X (formerly Twitter), with different text pre-processing techniques and classification models such as LDA-TFIDF and SVM-BOW. Their findings showed that the LDA-TFIDF model achieved second place with an accuracy of 92.4%, highlighting the significant improvement in sentiment analysis in Urdu, particularly with social media data, through better text representation models.

B. EMBEDDING TECHNIQUES IN URDU SENTIMENT ANALYSIS

The sentiment analysis study on Urdu text based on word embedding approaches first deployed, the context-based Urdu text emotion detection corpus (CUTEC) by [21]. The focus was put on the condition that sentiment analysis should be done considering the Urdu word embedding techniques. This method helps researchers build the connections between emoticons and Urdu language terms, as well as the field of sentiment analysis [22] which needs further exploration, this data serves as an important resource for the utilizing advanced learning models like RNN, LSTM, and bi-directional GRU, which can, in turn, lead to multiple iterations. A method description was provided by [17], detailing an emoji-based representation that employed the Siamese network architecture and Bi-LSTM structures. Emojis in the text will assist emoji2vec to understand the text better, allowing for the analysis of emotions in Urdu.

Furthermore, the study employed the Urdu language, as [23] studied, to examine people's sentiments as well as offensive and threatening content on social media platforms. The researchers generated an Urdu dataset that classifies Urdu tweets for threat recognition. It indicates how FastText embedding can improve multi-label classifiers. In another research by [24], they generated a dataset from the Facebook pages of Pakistan, applied different techniques for using the information, and developed a model that is more efficient than current models.

C. FUSION STRATEGIES IN URDU SENTIMENT ANALYSIS

The fusion method researched by [24] encompasses machine learning models such as the Naïve Bayes algorithm, Support Vector Machine, and deep learning frameworks including Convolutional Neural Network and Recurrent Neural Network, which resolves the difficulties of fusion methods in Urdu sentiment analysis. Reference [25] used the large SentUrdu-1M dataset which combined Urdu tweets and emojis. It is equipped with the training and test dataset. This dataset needs further consideration for the multiple ways in which the fusion strategies, like concatenation, neural network fusion, and attention mechanism fusion, contribute to the optimal and contextualized sentiment analysis framework for different inputs.

Reference [16] focused on with deep learning models for emotion detection and classification into various categories for the Urdu language. The Urdu Nastalik Emotion Dataset (UNED) was introduced, and models like machine learning and deep learning were compared, showing the distinct advantages of the latter [26] evaluated deep learning algorithms to demonstrate the efficiency of concept-based models. References [27] and [28] examined sentiment analysis in roman Urdu, [27] introduced a new dataset and used a CNN-LSTM network enhanced by Word2Vec embedding, which improved accuracy. Reference [28] developed a deep learning framework for online surveys in Urdu, which demonstrated the effectiveness of the R-CNN model in sentiment classification.

D. EVALUATION THE IMPACT OF EMBEDDING TECHNIQUES AND FUSION STRATEGIES

Integrating emoji2vec embeddings with traditional Urdu text embeddings and our proposed Urdu-Specific FastText emoji embedding models for Urdu language is an interesting analytical approach in this regard. Fusion methods are important techniques to help combine these resources together more effectively to analyze the accuracy of sentiment analysis. Combining different ways of fusion strategies, like concatenation fusion, using neural network fusion, and attention mechanism fusion, can help understand Urdu tweets and emoji features better [29].

The literature review elaborates that Urdu sentiment analysis has been improved by integrating emoji embedding and fusion techniques. However, there is a need for further research into the appropriate designs of these tools, which will promote Urdu language sentiment analysis and other languages of such nature and leading to a more specific understanding of digital-era sentiment analysis.

III. DATASET

As part of our research, we worked with a large dataset of 1.14 million Urdu tweets with 1,194 emojis [25], referred to as SentiUrdu-1M, to enhance the capability of natural language processing in the domain of Urdu. The goal was to minimize the complexities of Urdu tweet analysis [30]. The dataset is unorganized and contains emoticons, so we needed to explore how emotions are shown in a low-resource language [31]. Combing text and emojis refines the dataset and allows for a comparison of various data analysis methods [32], [33]. This method, therefore, not only improves the process of consuming the information, but also creates a very strong framework for comparing embedding strategies and tactics, as exemplified in Figure 1 of our data collection and preparation methodology.

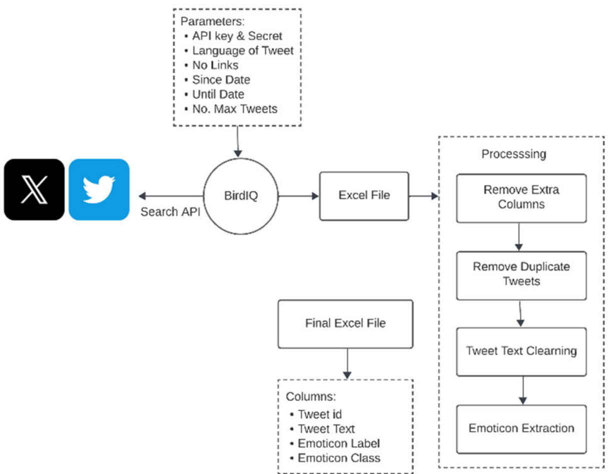

FIGURE 1. Workflow diagram of the tweet collection and preprocessing for Urdu sentiment analysis.

A. DATASET COMPOSITION

Every tweet in the SentiUrdu-1M dataset was meticulously annotated for accuracy and relevance in the context of the Urdu language. To ensure the credibility of the data, we avoided adding any external links. A wide range of emojis was strategically used as an intuitive sign of feelings [34]. Moreover, these emojis, which vary from well-known ones such as face with tears of joy '😂', to a wide spectrum of different feelings like joy '😊', grief '😢', fear '😱', surprise '😳', disgust '🤢', and anger '😡', have helped us to understand the deep emotional context integrated into social media communication [35].

Figure 2 depicts the length and emotional content of tweets in the Urdu-speaking X (Twitter) community. This assists us in comprehending significant emotional cues by analyzing the word count in tweets. Table 2 presents dataset characteristics comprising sentiment distribution, emoji usage, tweet length, and vocabulary size, with their implications for Urdu sentiment analysis.

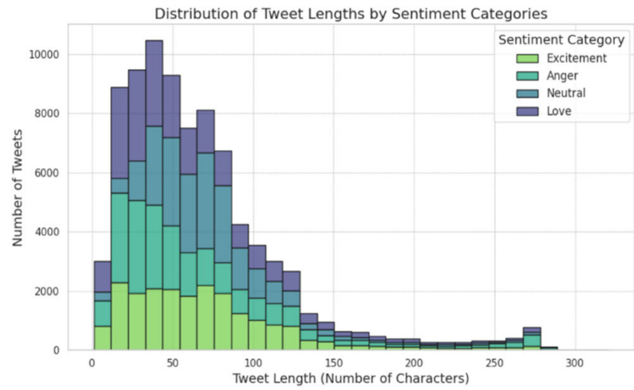

FIGURE 2. Distribution of tweets and sentiment classes in the SentiUrdu-1M Twitter X dataset.

TABLE 2. Summary of dataset characteristics including sentiment distribution, emoji usage, tweet length, and vocabulary size.

Statistic	Value
Total Tweets	1,019,418
Sentiment Distribution	
Excitement	373,561
Neutral	322,763
Anger	162,327
Love	160,767
Emoji Statistics	
Average Emojis Per Tweet	2
Minimum Emojis in a Tweet	1
Maximum Emojis in a Tweet	135
Most Common Emoji (😂)	577,449 occurrences
Second Most Common (❤️)	128,761 occurrences
Tweet Length Statistics (in words)	
Average	14.97

B. EMOJI SCORE-BASED ANNOTATION

For data annotation we used a point-based emoji annotation method which is designed to measure the sentiment of Urdu tweets in granular details, capturing even very subtle emotional subtleties [36]. This technique involves assigning emotional scores to emojis [37], which are context-dependent, thereby making sentiment classification better and enabling the transformation from the traditional methods. First, the Urdu language and NLP experts designed the emoji scores, and then the machine learning algorithms were used to enhance the scores' accuracy [38]. In Algorithm 1, tweets in Urdu are sorted into different categories like love, excitement, neutral, and anger by using scores from emojis [39].

Algorithm 1 Score-Based Sentiment Annotations for Urdu X (Twitter) Emojis

Objective: Using the emoji sentiment score as a classifier for Urdu language tweets for classifying them into sentiment classes.
Input: Corpus of Urdu X (Twitter tweets) with emojis.
Output: The most bet tweets category is about - Love, Excitement, Neutral, Anger

1: For each emoji, assign initial sentiment score (S) *by expert panel.*
2: Refine score (S) *based on context within corpus.*
3: For each tweet, compute aggregate score (A) *by summing refined scores* (S).
4: Categorize tweet based on A:
Love$(0.1 \leq A \leq 1)$, *Excitement* $(0.4 \leq A \leq 0.7)$, *Neutral* $(0.2 \leq A \leq 0.4)$, *Anger* $(A < 0.1)$.

C. DATA PREPROCESSING

During the preprocessing of the dataset, various methods were used as follows. Firstly, the text was converted to the Nastaliq script [30], the primary writing style in Urdu, which is also the most frequent practice of calligraphy in Urdu. Besides that, this enabled the researchers to overcome the linguistic difficulties of the language while also taking into consideration the visual and contextual elements, which play a vital role in accurate sentiment analysis. Moreover, we eliminated English letters to prioritize the use of Urdu and minimize the mixing of languages. URLs, hashtags, HTML labels, all punctuation, numbers, and white spaces were removed [40]. The primary concern was to correctly preserve the structure and context of tweets so that the tweet text and attached emoji conveyed the intended emotion correctly [41].

IV. METHODOLOGY

In this section, we outline the overall approach for sentiment analysis in Urdu digital communications, while recognizing the potential of emoji embeddings in Urdu tweet data.

We set two baseline models for comparison with proposed emoji embedding models to capture the linguistic and emotional complexity of Urdu [42]. Our first goal was to validate whether the pre-trained emoji2vec provides better results compared to our proposed novel Urdu-Specific FastText emoji embeddings that capture the sentiments conveyed through emojis within Urdu tweets. Secondly, we explored which fusion methods of Urdu tweet and emoji embeddings shows the best performance for improving the quality of sentiment analysis. The fusion methods used include concatenation, neural network fusion, and attention mechanisms.

A. BASELINE MODEL ESTABLISHMENT

We established the baseline models to evaluate sentiment analysis capabilities on only-text Urdu tweets [43]. These models, multilingual BERT (mBERT) and XLM-RoBERTa (XLM-R), are recognized for their effectiveness in Urdu natural language processing and provided a necessary foundation for assessing more complex sentiment expressions in Urdu [44].

In setting up these baseline models, it also emerged that their major drawback is a simplistic approach, as well as the fact that they were trained using only the textual data from Urdu tweets, while ignoring the importance of emojis [45]. Excluding emojis in these baseline models resulted in them missing the most socially defined types of emotions, which may be very important in a language as sensitive as Urdu. This prompted us to search for better methods to extend the utilization of emoji embeddings, aiming to provide increased intensity and polarity in sentiment analysis [46].

B. INTEGERATION OF EMOJI EMBEDDING TECHNIQUES

To address the limitations identified in the baseline models, we explored the integration of two emoji embedding techniques: emoji2vec and the proposed Urdu-Specific FastText emoji embeddings. While the use of emojis was found to be effective in sentiment analysis in general, we hypothesized that coupling Urdu text with emoji embeddings would further improve the accuracy of sentiment analysis.

1) EMOJI2VEC EMBEDDING FOR EMOJIS

Some challenges that relate to emojis in Urdu sentiment analysis might include a failure to recognize emojis multiple aspects of semantics as rule-based semantic networks may fail to address them. To achieve this, emoji2vec employs the use of emoji symbols exclusively to provide more substantial representation for each emoji other than just association with a particular word [47]. This specific approach to the research is important due to the fact that we consider the vivid representation of emotion as a distinguishing feature of the Urdu language. Emoji2vec [13] performs the process of transmutation of emojis to their Unicode descriptions to assure they share a single meaning and improve the accuracy in detecting sentiments.

2) URDU-SPECIFIC FASTTEXT EMBEDDINGS FOR EMOJIS

These pre-trained traditional models, such as emoji2vec, are advantageous for analyzing the general meaning of emoji [48]. However, they may be inaccurate in capturing the true semantics and range of usage in Urdu. To address this issue, our study tried to overcome this by using Urdu emojis with their own Urdu-specific FastText emoji embeddings. In this way, we aim to represent the language complexity in Urdu. This method is highly significant in sentiment analysis, and it is successful in solving the problems of the Urdu language, which possess depth and details. FastText [49], which is characterized by morphological subtlety, is the foundational algorithm used in our approach. Unlike other models that generally identify words or sentence fragments, our algorithm treats the sentences as bags of character n-gram, allowing us to have an in-depth analysis of the real components. Our approach includes the translation of emojis into descriptive narratives of Urdu rather than just sense and sentiment. In Equation (1) given the emoji e, it is turned into description d by a process known as $d = demojize(e)$ then they are sent to FastText, where each catalog entry is adjusted into a set of character n-grams

$$F(d) = \sum\nolimits_{g \in G(d)} Embed(g) \tag{1}$$

This process produces Urdu-Specific FastText emoji embeddings that are both sensitive to the linguistic peculiarities of Urdu and semantically rich. As Figure 3 shows, there is a step-by-step transition from emoji translation to n-gram vectorization, which shows the depth of our framework from different perspectives. Vector map contrasts emoji2vec and Urdu-Specific FastText vector embedding method for emojis "❤️" and "😂". Figure 4 visualizes the capacity of a strategy to differentiate the semantics and sentiment of both emojis [50]. Urdu-Specific FastText emoji embeddings have captured the characteristics, especially morphological elements of Urdu, which are, unlike emoji2vec, general-purpose by nature.

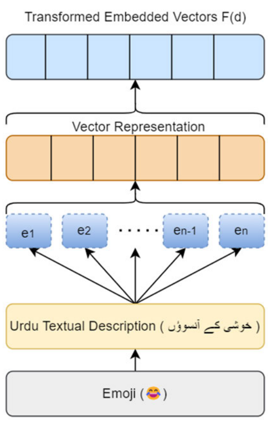

FIGURE 3. Process flow of generating Custom FastText Embeddings from emojis for Urdu sentiment analysis.

Our study shows the significance of choosing the embeddings that recognize the lingual context meaning. Hence, the Urdu-Specific FastText is an efficient way to understand the real meaning of Urdu emojis.

C. FUSION STRATEGIES FOR TEXT AND EMOJI EMBEDDINGS

Our second objective was to combine these emoji embeddings using three fusion strategies. This integration of different input modalities is the key aspect for understanding multimodal sentiment analysis frameworks. To obtain this, we conducted three state-of-the-art fusion techniques, each

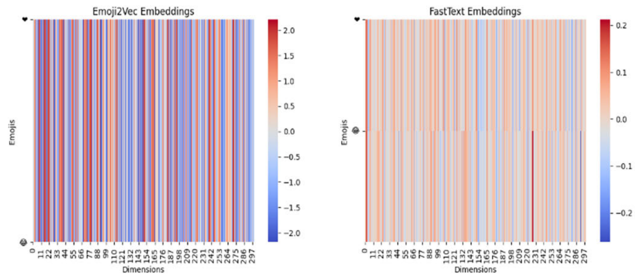

FIGURE 4. Heatmap comparison of vector representations for emojis "❤️" and "😂" between Emoji2Vec and Urdu-Specific FastText emoji embeddings.

brings up unique contributions to the sentiment analysis technique.

1) CONCATENATION FUSION

Concatenation fusion is the basic type of the fusion strategy and an integral part of its effective implementation, ensuring that the depth expressed by the tweets and the clarity of the emotions conveyed by the emojis are cohesively integrated [51]. Consolidation implies that tweets and emoji encodings are respected individually because the characteristics of each are maintained and integrated into a unique vector. This is achieved by combining the tweets T and emojis E (which have dimensions d_t and d_e respectively) into a single concatenated vector C. Through the resulting concatenated vector C, the dimensions of text embedding are d_t and emoji embedding of d_e, thus combining all into a new vector with a dimensionality of $d_t + d_e$, graphically represented in Figure 5 to ensure balanced integration of both text and emoji features in the sentiment analysis process.

For a language culturally rich as Urdu, the concatenation approach is essential. It ensures that emojis, along with the intricate sentiment expressions encoded in Urdu text, are preserved in the fused embeddings without error. While this method acts like a solid reference point for contrasting more complicated fusion techniques, it remains significant in maintaining the authenticity of multimodal sentiment signals in Urdu sentiment analysis.

2) NEURAL NETWORK FUSION

Concatenation performed better initially, however, neural network fusion method which merges Urdu language text with emojis as illustrated in Figure 6, needs careful consideration.

Neural network fusion employs a deep learning technique to do both the analysis and the embedding of the Urdu tweets and emojis, discreetly exposing the connections within the data that would otherwise remain undetected [52]. This approach relies on a multi-layer perceptron (MLP) to find non-linear interactions within the data. This function takes the text embedding T and the emoji embedding E as inputs and, via a sequence of weighted neural network layers provided by parameters θ, generates a fused embedding F using

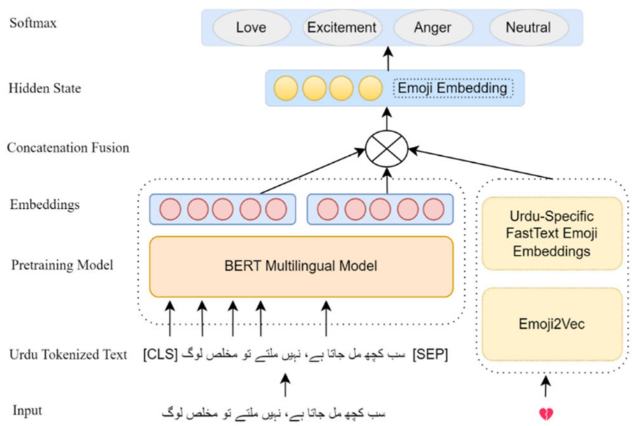

FIGURE 5. Illustration of the concatenation fusion strategy for integrating Urdu Tweet and Emoji Embeddings.

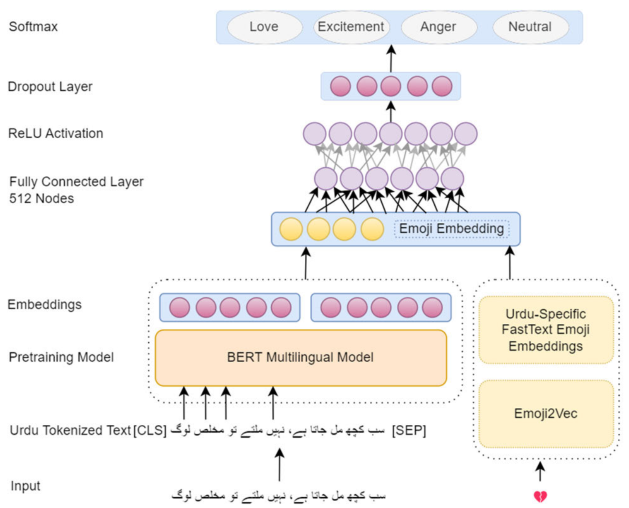

FIGURE 6. Schematic representation of the neural network fusion strategy for Urdu tweet and Emoji embeddings.

Equation (2).

$$F = f\,(T, E; \theta) \tag{2}$$

3) ATTENTION MECHANISM FUSION

We have gone beyond typical fusion algorithms by employing attention mechanism fusion to conduct context-sensitive sentiment analysis. It helps us understand how emotions are influenced by context. Like the human brain, this mechanism carefully analyzes specific aspects of the input data, such as Urdu tweets and emojis, that are most relevant to the emotion being conveyed. The adjustment where specific instances are presented is very important for a more generalized comprehension of sentiment, which is shown in Figure 7, Attention fusion involves finding out the sentences and corresponding emojis that are close to the context and interact deeply with the overall sentiment in Urdu tweets sentiment analysis [53]. The attention mechanism calculates attention scores a that indicate each slice's relevance to the sentiment analysis task. These weights are estimated by a scoring function which is a measure of the match and relevancy between embedding states H that represent our encoder and a decoder state S as shown in Equation (3).

$$a_i = \frac{exp\,(score\,(S, H_i))}{\sum_j exp\left(score\left(S, H_j\right)\right)} \tag{3}$$

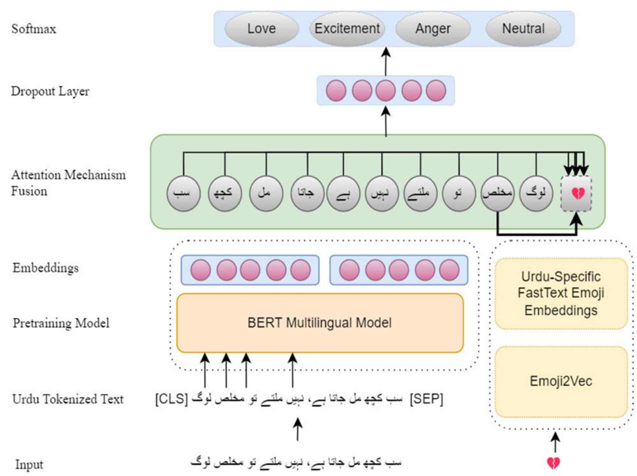

FIGURE 7. Attention mechanism fusion for Urdu Tweet with Emoji on sentiment analysis.

The above formula makes sure that each segment's level of importance to the combined representation is weighted by its contextual significance, this helps the model to pay more attention to information that is most relevant to the sentiment being analyzed. The attention mechanism, which is dynamic in nature, assures that this merging process is not only contextually sensitive but also a truly diverse one. It can be merged with various other fusion methods like neural network architectures, which improve the sentiment analysis model's capability to extract the sublime and complex cues of sentiment from versatile and varied Urdu communication.

D. MODEL TRAINING FRAMEWORK AND HYPERPARAMETER OPTIMIZATION

Data was initially divided into separate training, validation, and testing subsets to achieve complete evaluation of model performance. We assigned 80% of data for training, allowing the model to learn from a large and diverse variety of examples. The rest of the data was distributed evenly between the validation set and the test set, with 10% each to allow the model, adjustment, and final model performance.

The training process leveraged stochastic gradient descent (SGD) and adaptive optimization methods to iteratively refine model parameters, ensuring convergence to optimal solutions. The choice for hyperparameters was affected by both an empirical method of testing actual data and a reference to existing literature. This iterative procedure was aimed at achieving complexity and generalization, with the configurations, summarized in Table 3.

TABLE 3. Hyperparameters and configuration settings for sentiment model training.

Hyperparameter Configuration	Value
Batch Size	64
Training Data Split	80%
Validation Data Split	10%
Testing Data Split	10%
Sequence Length (tokens)	150
Learning Rate	0.0001
Epochs	10
Emoji Embedding Size (dimensions)	300
Hidden Layer Size (units)	128
Dropout Rate	0.5
Number of Classes	4

E. EVALUATION MEASURES

The evaluation measures used for our Urdu sentiment analysis model, which are employed to fine-tune the multilingual BERT model to the Urdu tweets, consist of a comprehensive set of metrics which can reflect the interaction between text and emojis. These metrics are vital ways for evaluating the quality of our models, considering their capabilities to correctly identify the sentiments.

1) ACCURACY

This statistic reflects the total accuracy of the model and is calculated as an overall ratio by summing up the correctly classified data and diving it by the total data. In mathematical terms, presented as Equation (4):

$$Accuracy = \frac{True\ Positives + True\ Negatives}{Total\ Observations} \tag{4}$$

2) PRECISION

The length of the measure shows how accurately the model can predict the true instances, which is significant in scenarios where the cost of false positives is very high, as it is important to accurately predict the true instances. It is defined in Equation (5):

$$Precision = \frac{True\ Positives}{True\ Positives + False\ Positives} \tag{5}$$

Considering the vast hidden features of emotions in Urdu, our model can correctly identify positive feelings counted acceptable, as it is trustworthy with high precision.

3) RECALL

We use this key metric to determine whether we are detecting all positive expressions. The detection process's accuracy is determined. It is calculated as follows in the Equation (6) below:

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives} \tag{6}$$

Unlike non-emotional content, feelings also play an important part in how we communicate; therefore, in the domain of Urdu sentiment analysis, especially when considering expressions that are harder to understand, the high recall is crucial.

4) F1-SCORE

The F1-score helps to mitigate the imbalance between precision and recall that would otherwise lead to a trade-off, and therefore F1-score is significant evaluation metric as defined in Equation (7). It is an average of two measures; precision, on the other hand, is recall.

$$F1Score = 2 \times \frac{Precision \times Recall}{Precison + Recall} \tag{7}$$

The precision-recall trade-off is consequently an important technical metric, especially at this level of our research, as it may further prove the balance of our model in detecting the most user appropriate sentiment of Urdu.

5) CONFUSION MATRIX

Confusion matrix is very important for identifying how a model handles all the individual classes about the Urdu and the use of the emoji [54].

Those metrics helped us draw a compressive conclusion on sentiment analysis models' performance evaluated based on the overall accuracy and balance between recall and precision. Such a highly developed evaluation strategy is necessary for Urdu social media communication, which is linguistically rich, as far as sentiment analysis is concerned.

V. RESULTS

In this section, we present the findings of both the baseline and proposed models of emoji embeddings used in this research. We compare the performance of the baseline model with the proposed model. Additionally, we conduct a comparison analysis of fusion strategies using concatenation, neural network fusion and attention mechanism fusion.

TABLE 4. Performance evaluation of multilingual bert (mBERT) and xlm-robert (XLM-R) on urdu text-based data.

Model	Accuracy	Precision	Recall	F1 Score
mBERT	64.07	62.79	64.07	62.63
XLM-RoBERTa	65.22	64.89	65.22	64.21

A. BASELINE MODEL EVALUATION

The results, summarized in Table 4, reveal the performance of the baseline models multilingual BERT (mBERT) and XLM-RoBERTa (XLM-R) models across several key metrics.

Although these models are capable of processing multilingual content, their performance on the Urdu dataset highlighted inherent limitations in capturing the nuanced

TABLE 5. Performance metrics of different fusion strategies for emoji2vec embedding.

Fusion Strategy	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	Loss
Concatenation	87.67	91.80	87.86	87.03	0.215
Attention	88.93	91.83	87.92	87.19	0.195
Neural Network	87.90	91.85	87.90	87.19	0.159

emotional content typical of Urdu tweets, underscoring the need for enhanced modeling techniques.

B. EMOJI EMBEDDINGS

To address these limitations, we explored the integration of emoji embeddings with Urdu tweet data, hypothesizing that emojis significantly enrich the sentiment analysis by adding emoji embedding layers of emotional expression not captured by text alone.

1) EMOJI2VEC EMBEDDINGS ANALYSIS

We employed emoji2Vec embeddings in conjunction with three fusion techniques to better capture the sentiment nuances in digital communications in Urdu. Our comparative analysis involved three primary fusion strategies:

Concatenation Fusion: This fusion method achieved an accuracy of 87.67%, illustrating the utility of combining textual and emoji data linearly to provide a more holistic sentiment analysis, the confusion matrix shown in Figure 8a, characterizes in detail the performance of our model in sentiment classification.

Attention Mechanism: This fusion provided a slight improvement with 88.93% accuracy, highlighting the effectiveness of dynamically focusing on more relevant segments of the data, The depiction of this process is shown in Figure 8b, measured metrics demonstrate the advanced capabilities to grasp the intricate connection between tweets and emojis as opposed to concatenation, which scores a little higher in terms of accuracy and precision.

Neural Network Fusion: This method of fusion demonstrated a significant enhancement, with an accuracy of 87.90%, showcasing the depth and adaptability of neural networks in integrating complex data types for sentiment analysis. The confusion matrix for neural network fusion shown in Figure 8c provides a detailed picture of its performance with respect to each class.

The detailed performance metrics of these fusion strategies are outlined in Table 5, which includes accuracy, precision, recall, F1 score, and loss for each method. These metrics further substantiate the efficacy of each fusion technique, with the attention mechanism fusion not only achieving the highest accuracy but also demonstrating considerable improvements in precision and recall, closely followed by neural network fusion, which shows the lowest loss, indicating a robust model performance.

C. URDU-SPECIFIC FASTEXT EMOJI EMBEDDINGS ANALYSIS

Building upon the emoji2Vec findings, we introduced novel Urdu-Specific FastText emoji embeddings, tailored specifically to the linguistic intricacies of Urdu. This approach was evaluated using the same three fusion techniques, yielding notably superior results:

Concatenation Fusion: The ability of this strategy to align with the linguistic uniqueness of Urdu language is shown graphically in the confusion matrix in Figure 9a, which highlights its functionality to distinguish and classify sentiments in accordance with their nuances, Achieved an accuracy of 89.20%.

Attention Mechanism Fusion: The corresponding confusion matrix shown in Figure 9b provides a detailed view of this strategy's precision, achieved an accuracy of 90.10%.

Neural Network Fusion: Achieved the highest accuracy at 95.00%. Figure 9c is a visual depiction of the Urdu-Specific FastText emoji embeddings in unison with the advanced neural architecture enhances their effectiveness through the creation of synergistic potential. The performance metrics for these strategies, including accuracy, precision, recall, F1 score, and loss, are comprehensively detailed in Table 6. These findings support the viability of the suggested Urdu-Specific FastText emoji embeddings, as well as the superiority of the neural network fusion strategy to enhance the performance of the model.

VI. COMPARATIVE ANALYSIS AND INSIGHTS

In this section, we closely examine the comparison of various emoji embedding techniques and fusion strategies in our study, highlighting their strengths and limitations. Through the comparison of our proposed emoji embedding technique and baseline multilingual models, we gain valuable insights into improving sentiment analysis for Urdu tweets.

A. BASELINE COMPARISON

This is evident in the baseline results, which show that multilingual models have challenges understanding sentiments in Urdu tweets. The performance of the Urdu- Specific FastText emoji embeddings model significantly improved the predictive capability, thus establishing the significance of adopting specialized model for low-resource languages.

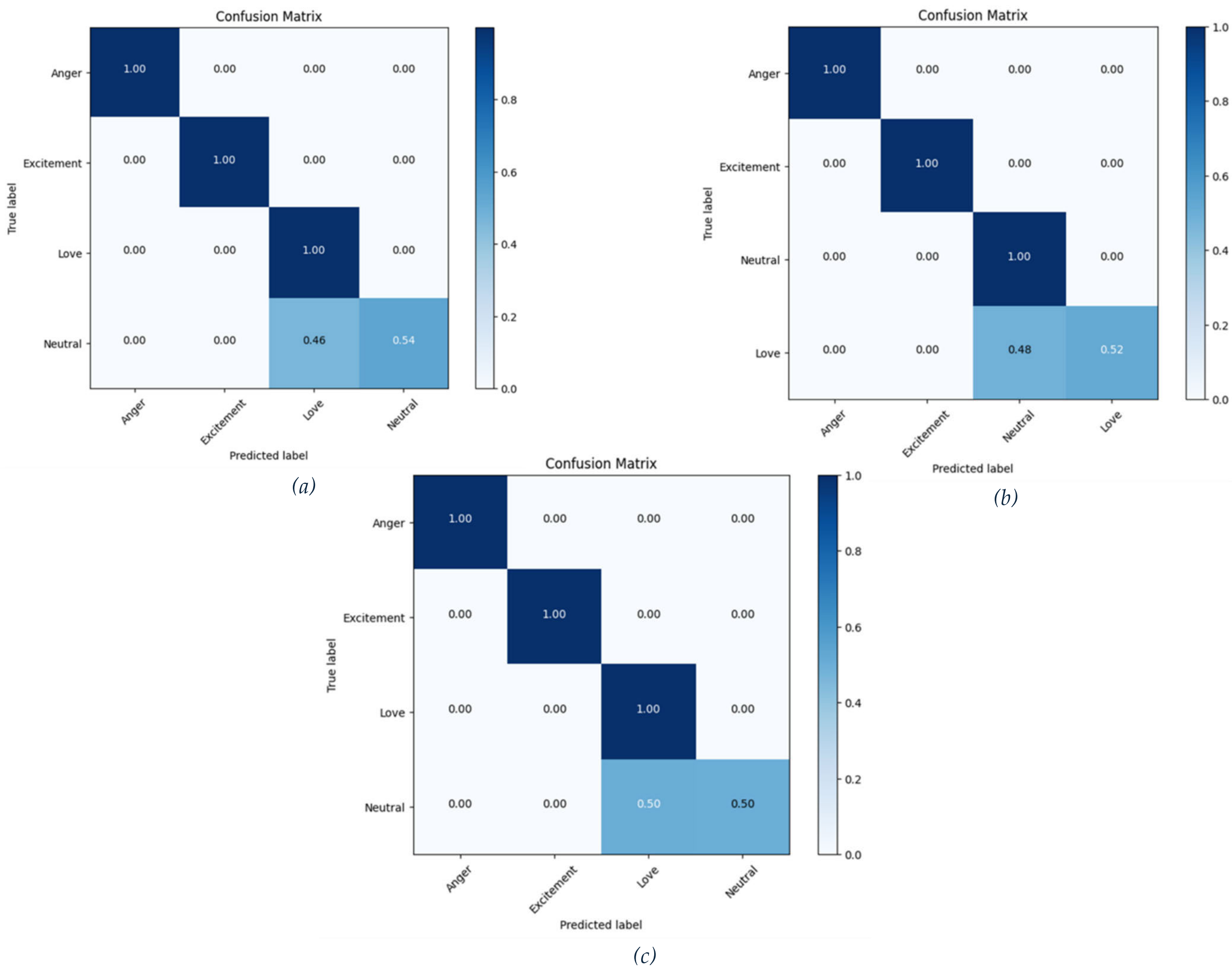

FIGURE 8. (a) Confusion matrix for concatenation strategy using Emoji2Vec Embeddings, (b) Confusion matrix for attention mechanism fusion strategy using Emoji2Vec Embeddings, (c) Confusion matrix for neural network fusion strategy using Emoji2Vec Embeddings.

TABLE 6. Performance metrics of urdu-specific fasttext emoji embeddings analysis.

Fusion Strategy	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	Loss
Concatenation	89.20	92.00	89.50	88.95	0.2345
Attention	90.10	92.55	90.45	89.98	0.2201
Neural Network	**95.00**	94.81	94.49	94.52	0.2717

The results reported in Table 7, not only surpassed the emoji2vec embedding strategies but also dramatically improved upon the text-only baselines, confirming the profound impact of culturally and linguistically specific embeddings in sentiment analysis.

To synthesize the data derived from emoji2vec and its application with three different fusion strategies, we refer to Figure 10, which depicts the comparison of performance metrics. This aggregated representation helps to get a clear view of each strategy's own strengths and weaknesses, and thus informs future methodological choices in the sentiment analysis of digital communications in Urdu.

This visual representation aids in understanding the effectiveness of each strategy across multiple performance metrics: accuracy, precision, recall, and F1-score. The graph shows all three strategies being around the same in accuracy where the neural network model slightly edges the others, hinting that adding complex design can marginally

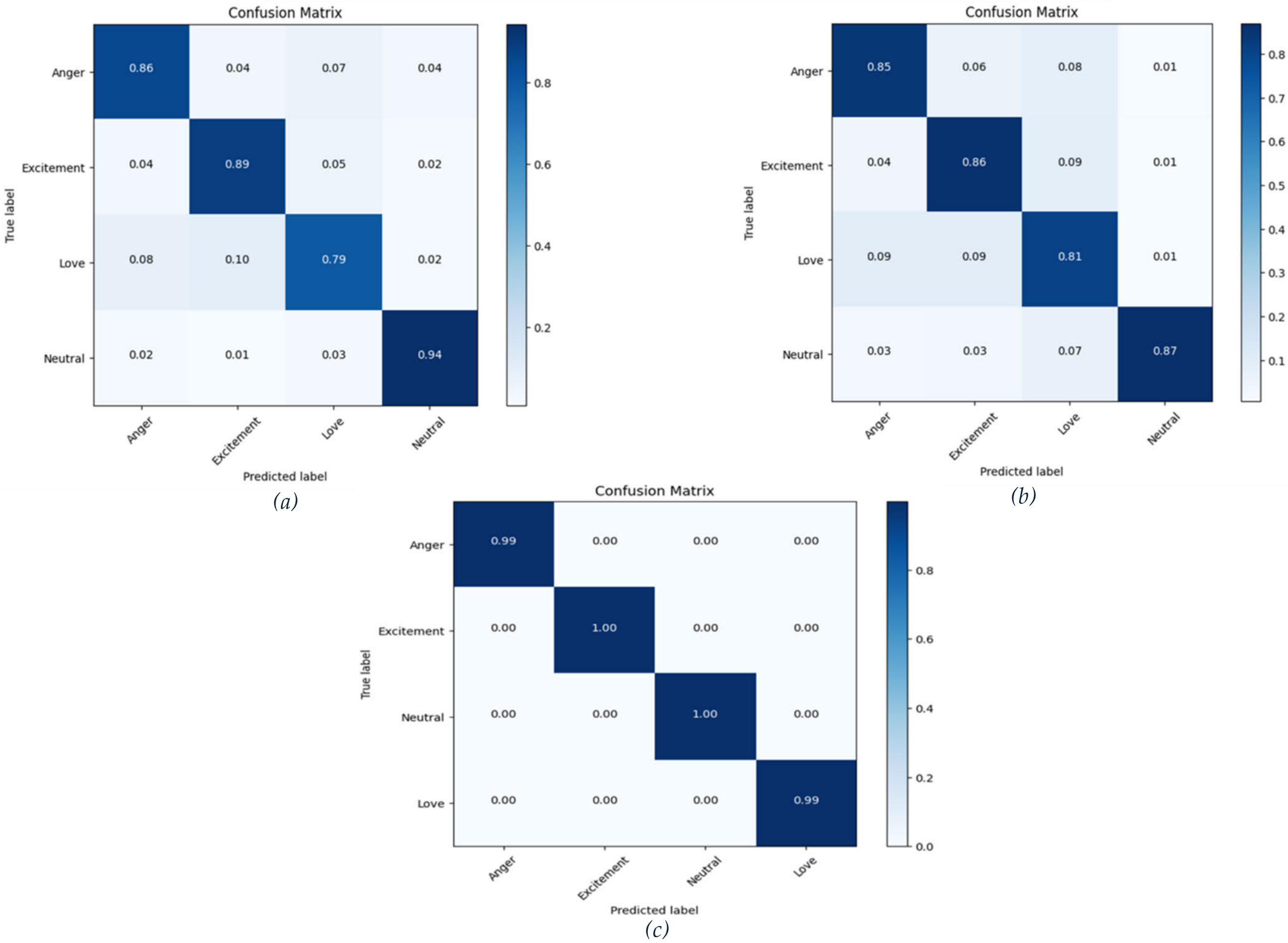

FIGURE 9. (a) Confusion matrix for concatenation fusion strategy using Urdu-Specific FastText Emoji Embeddings, (b) Confusion matrix for attention mechanism fusion strategy using Urdu-Specific FastText Emoji Embeddings, (c) Confusion matrix for neural network fusion strategy using Urdu-Specific FastText Emoji Embeddings.

improve the correctness rate. The attention mechanism fusion strategy is just slightly more accurate, indicating its impact on adequately filtering positive cases without resulting in many false positives. The recall rates are the same for all strategies, indicating that each approach identifies all crucial information equally. Lastly, the F1 score is almost the same for all strategies, therefore, there is a balanced tradeoff between precision and recall for each technique.

B. ANALYSIS OF FUSION STRATEGIES USING URDU-SPECIFIC FASTTEXT EMOJI EMBEDDINGS

The complete comparison of the results elicited from the use of various fusion strategies utilizing the Urdu-Specific FastText emoji embeddings is given in Figure 11. The graph clearly shows that neural network performs better than other methods in all measures, demonstrating its high capacity to consider the linguistic specificity of the Urdu tweets through the unique emoji embeddings.

The neural network algorithm can deliver the highest accuracy and F1 score among all strategies proving that it is the best in the class in that it is able to achieve a good balance between precision and recall as well. On the other hand, the attention mechanism, although a little behind the best model, displays good accuracy, thus illustrating its capability

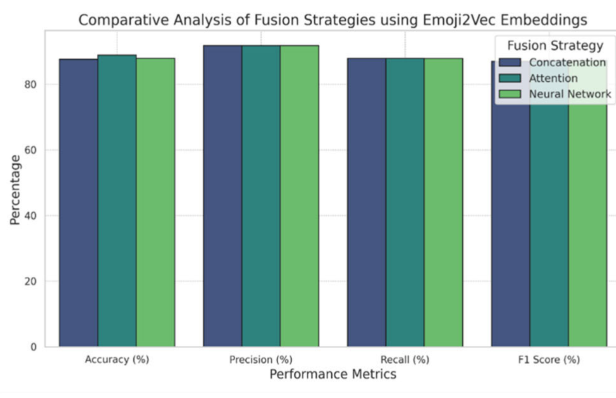

FIGURE 10. Performance comparison of fusion strategies using Emoji2Vec Embeddings.

TABLE 7. Performance metrics comparison of baseline models and advanced emoji embedding strategies.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
mBERT (Baseline Model)	64.07	62.79	64.07	62.63
RoBERTa (Baseline Model)	65.22	64.89	65.22	64.21
Concatenation (Emoji2Vec)	87.67	91.8	87.86	87.03
Attention (Emoji2Vec)	88.93	91.83	87.92	87.19
Neural Network (Emoji2Vec)	87.9	91.85	87.9	87.19
Concatenation (Urdu-Specific FastText)	89.2	92.00	89.5	88.95
Attention (Urdu-Specific FastText)	90.1	92.55	90.45	89.98
Neural Network (Urdu-Specific FastText)	95.00	94.81	94.49	94.52

of more selectively focusing on the relevant features in the data. The concatenation strategy was revealed as a highly generic approach but less dynamic one than the more complex models, reaching average performance.

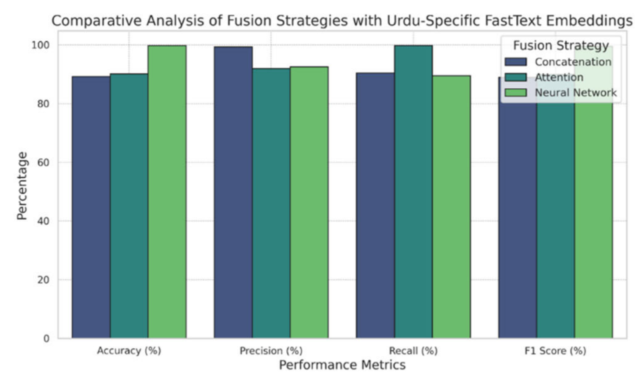

FIGURE 11. Performance comparison of fusion strategies using Urdu-Specific FastText Emoji Embeddings.

C. EMOTIONAL NAUNCES IN URDU AND EMOJI REPRESENTATION

Emojis these days often called as the modern fonts that portray emotions, thoughts, and feelings well. The introduction of these symbols into Urdu is expected to be a bridge linking classical literary architecture and contemporary digital representation. It is evident from our selected dataset that some emojis are dominant. For example, '❤️ میری ادا بھی ہے نمودعشقوہ میری آواز میرا لہجہ', with translation is 'He is my voice, my tone, my style, a symbol of love', contains the emoji '❤️'. The unambiguous connection between romance and love in the tweet is further consistent with the text's mood. Moreover, issues arise with sentences containing more complex meanings. Due to the emotional depth of words in Urdu, it is difficult for the embeddings to render their meaning completely. Another example of this sentence in Urdu '💔 میرے بعد کس کو ستاؤ گے', translated as 'Who will you torment after I'm gone?' with its corresponding emoji '💔'. The raw feelings of hurt or sadness are clearly present, but what is more interesting are the deeper feelings under them, such as rejection, desire, and sarcasm. This is where traditional embeddings may fail.

D. DIVERSITY OF FUSION STRATEGIES

During our research on various fusion techniques, we obtained different results. Some of them were very good at expressing a particular emotion, while others struggled attempting to represent the complexities of the various sentiments. We have a real example from our dataset, "ہیں‌لوگ اب ناراض نہیں بیزار ہو جاتے," which translates to "People don't get disillusioned now; instead, they are neutral," was paired with the emoji "💔". However, the less noticeable "angry" or "heartbreak" sentiment might have been missed if the more subtle "indifference" or "resignation" were not included. It is very important to comprehend that various strategies of fusion bring different strengths and weaknesses to the table.

VII. CONCLUSION & FUTURE WORK

The role of emoji usage in Urdu sentiment analysis has not yet been explored. Our research objective was to highlight the importance of adding emoji embeddings to Urdu tweets to enhance sentiment analysis in this low-resource language. Our only Urdu text-based baseline models, multilingual BERT (mBERT) and XLM-RoBERTa (XLM-R), showed limitations in capturing the complete meaning of Urdu tweet sentiments; they required additional features to improve accuracy. After performing various experiments based on analyzing the performance of pre-trained emoji2vec and Urdu-Specific FastText emoji embeddings, along with fusion strategies for integrating Urdu tweets and emoji embeddings, our proposed approach Urdu-Specific FastText emoji embeddings significantly improved accuracy, especially when using the neural network fusion strategy, achieving 95% accuracy. These enhancements not only improved accuracy but also surpassed our baseline-only Urdu text-based models. Our research demonstrates that using emojis helps researchers understand context more effectively than relying solely on text. In the future, we aim to expand our research to other low-resource languages, exploring whether our Urdu-specific sentiment analysis techniques can be adapted to similarly complex linguistic contexts. This expansion will help in broadening the applicability of our methods and potentially improving sentiment analysis frameworks across diverse linguistic landscapes.

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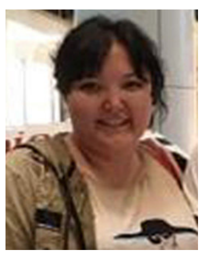

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